

1 Summary

Model (Run-name): model1 (test)

Iteration: 9

TAC: 154200.20534662405

Took: 2.254619836807251s

Total Time: 9.410642147064209s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

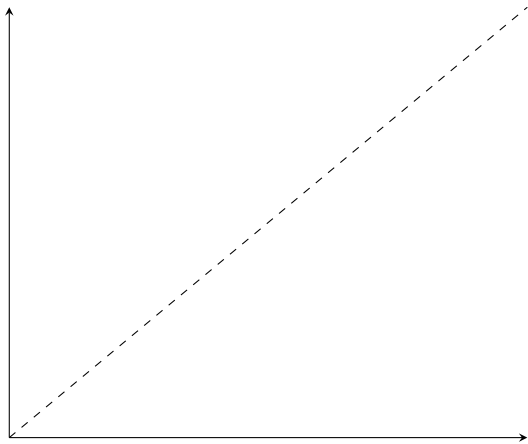
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

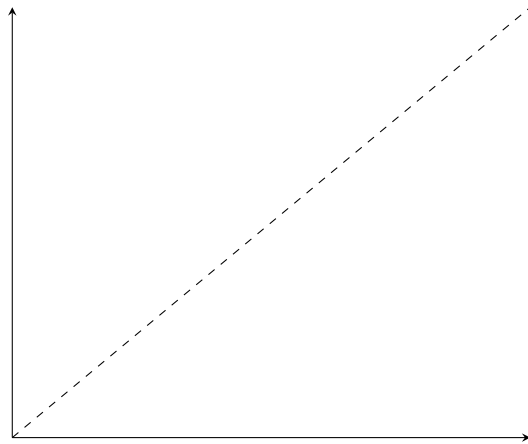
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.050880792740000896	$b_{1,1,1}^{H,out}$	955.6219305874799
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.019140932920041452	$b_{1,2,2}^{H,out}$	3826.826647122988
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.025472919887073416	$b_{2,1,2}^{H,out}$	3977.9035870602656
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.015073369208121666	$b_{1,1,1}^{C,in}$	8254.942916184675
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.0094646152736325296	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.0222626104816614736	$b_{2,1,2}^{C,in}$	4173.751008635552
$A_{1,1,1}^\beta$	71.22933662920877	$q_{1,1,1}$	723.17335287701196	$b_{1,1,1}^{C,out}$	8978.116269061686
$A_{1,2,2}^\beta$	74.6496383881616	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	118.88765652962957	$q_{2,1,2}$	2365.2399706844826	$b_{2,1,2}^{C,out}$	6538.990979320034
$A_{cu,1}^\beta$	2.5450346292088537	$q_{cu,1}$	126.8266471229881		
$A_{cu,2}^\beta$	38.21652191748154	$q_{cu,2}$	2034.7600293155178		
$A_{hu,1}^\beta$	13.468285806082731	$q_{hu,1}$	511.5866764385062		
$A_{1,1,1}$	71.22933662920877	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	74.6496383881616	$t_{1,2}^{(H)}$	577.6826647122989		
$A_{2,1,2}$	118.88765652962957	$t_{1,3}^{(H)}$	382.6826647122988		
$A_{cu,1}$	2.5450346292088537	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	38.21652191748154	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.468285806082731	$t_{2,3}^{(H)}$	471.7380014657759		
$dt_{1,1,1}$	34.105778429233744	$t_{1,1}^{(C)}$	615.8942215707663		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	567.6826647122988		
$dt_{1,2,2}$	77.6826647122989	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	32.68266471229881	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	22.317335287701198	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	61.73800146577591	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	62.68266471229881	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	151.73800146577588	$t_{1,2,2}^H$	382.6826647122988		
$dt_{hu,1}$	64.10577842923374	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	2.5827619745607566	$t_{1,1,1}^C$	617.507387405553		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	10.751090775838556	$t_{2,1,2}^C$	636.8366519389214		
$f_{1,1,1}^C$	14.539285605607231	$b_{1,1,1}^{H,in}$	1678.7952834644918		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5776.826647122988		
$f_{2,1,2}^C$	10.1798805088672	$b_{2,1,2}^{H,in}$	6343.143557744748		

3 *RecLMTD* Tangent Points

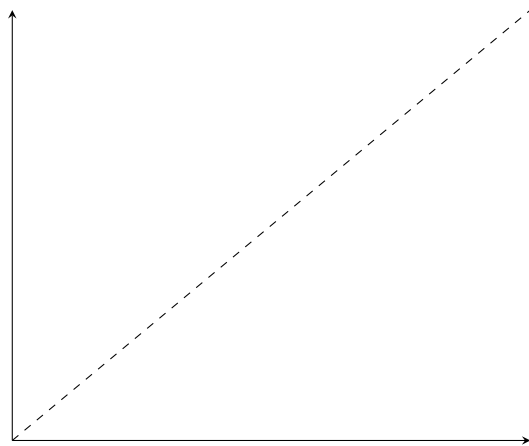
- $(1, 1, 1)$
 - Value: 0.050881
 - Error: 0.000014959742912
 - Relative error: 0.000293929103757
- $(1, 2, 2)$
 - Value: 0.019141
 - Error: 0.000098785322057
 - Relative error: 0.005134447439073
- $(2, 1, 2)$
 - Value: 0.025473
 - Error: 0.000339325399461
 - Relative error: 0.013145907908976
- $cu, 1$
 - Value: 0.015073
 - Error: 0.000015499644356
 - Relative error: 0.001027223744054
- $cu, 2$
 - Value: 0.009465
 - Error: 0.000000506250257
 - Relative error: 0.000053485869704
- $hu, 1$
 - Value: 0.022263
 - Error: 0.000001567621734
 - Relative error: 0.000070410042824



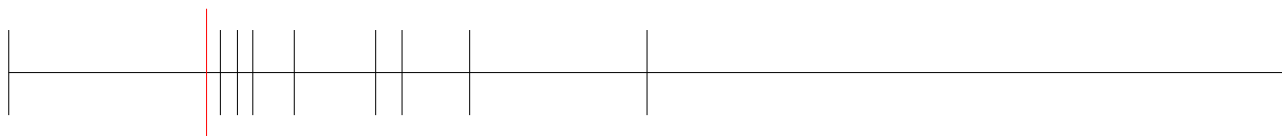
(a) *RecLMTD* - (1, 1, 1)



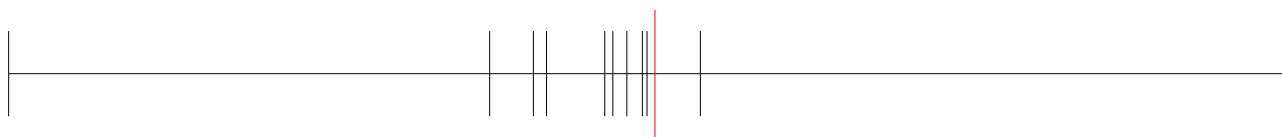
(b) *RecLMTD* - (1, 2, 2)



(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1678.7952834644918
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5776.826647122988
 - Absolute error: 9.094947017729282e-13
 - Relative error: -1.5743846186312003e-16
- (2, 1, 2)
 - Value: 6343.143557744748
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 955.6219305874799
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3826.826647122988
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 3977.9035870602656
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8254.942916184675
 - Absolute error: 1.2425205803920107
 - Relative error: 0.00015054103260808287
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4173.751008635552
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 8978.116269061686
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6538.990979320034
 - Absolute error: 56.06995891476345
 - Relative error: 0.008648872743980816



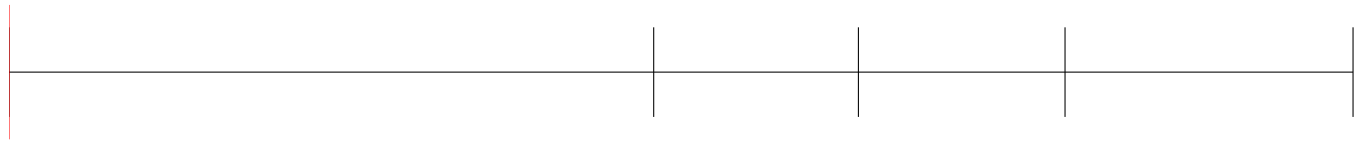
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



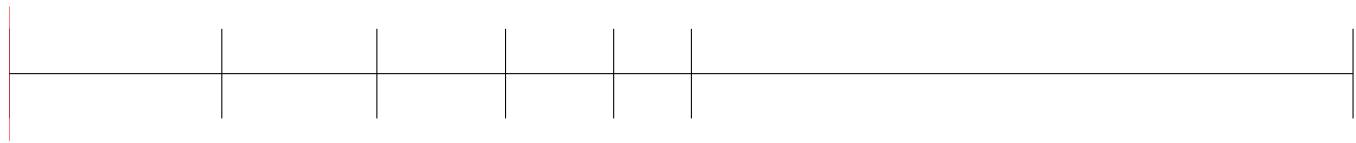
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



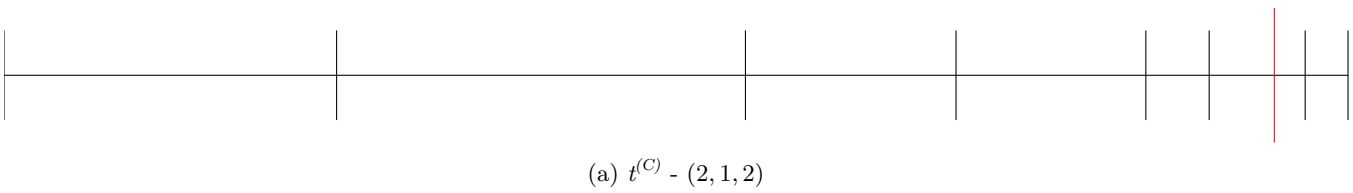
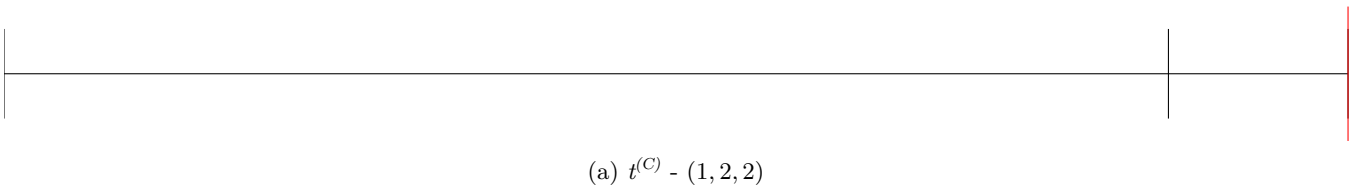
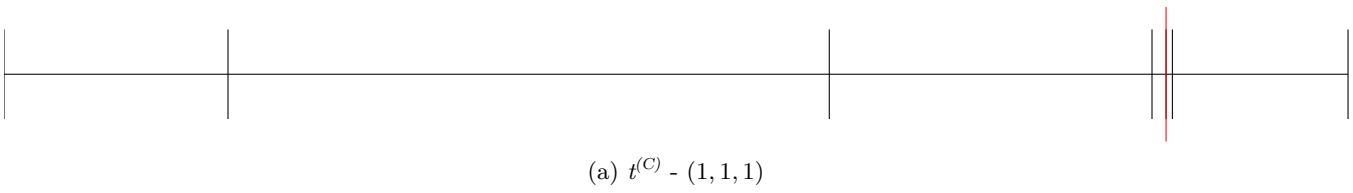
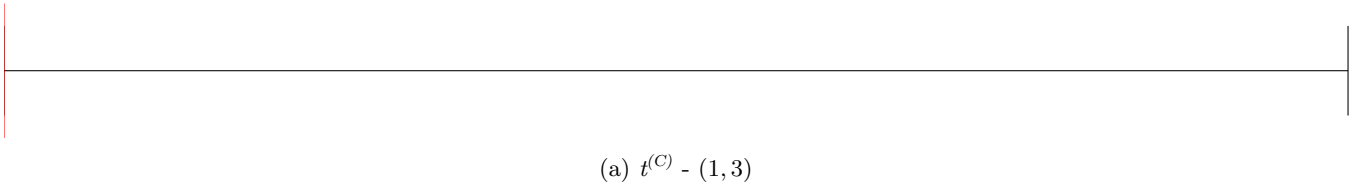
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
 - Value: 71.229337
 - Error: 2.361930336444757
 - Relative error: 0.032095253062393
- $(1, 2, 2)$
 - Value: 74.649638
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 118.887657
 - Error: 1.611480044269783
 - Relative error: 0.013373374200748

5.2 A_{CU}

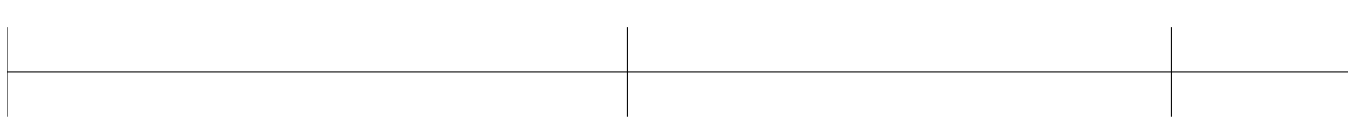
- $cu, 1$
 - Value: 2.545035
 - Error: 1.278375125817067
 - Relative error: 0.334354727252716
- $cu, 2$
 - Value: 38.216522
 - Error: 0.299919785791701
 - Relative error: 0.007786798897527

5.3 A_{HU}

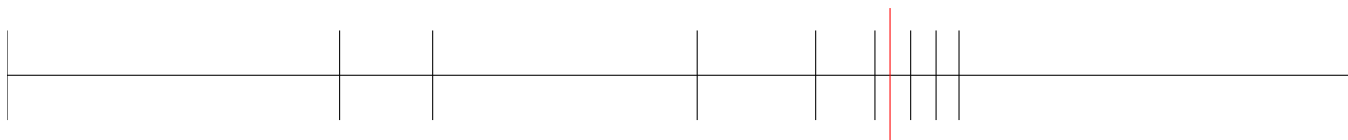
- $hu, 1$
 - Value: 13.468286
 - Error: 0.198820080107161
 - Relative error: 0.014547343216830



(a) $A^\beta - (1, 1, 1)$



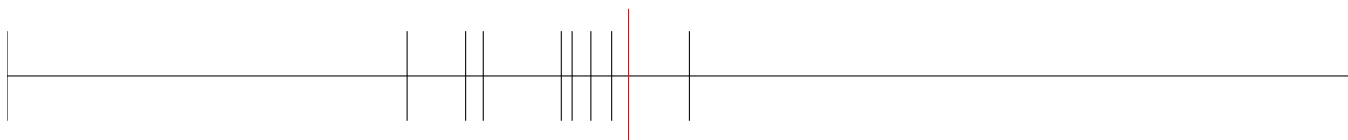
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

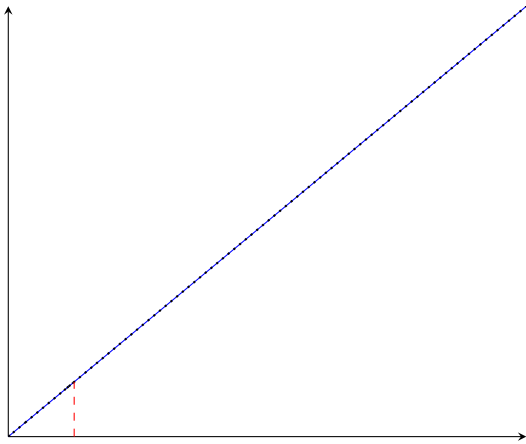


(a) $A^\beta - hu, 1$

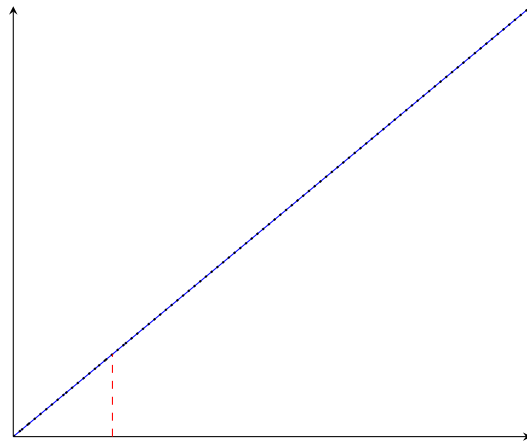
6 A^β Breakpoints

- (1, 1, 1)
 - q : 71.229337
 - $\hat{q}^\beta$: 71.229337
 - q^β : 71.229337
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 69.14107665413653, 69.89902342903967, 70.30181758862284, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 74.649638
 - $\hat{q}^\beta$: 74.649638
 - q^β : 74.649638
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 69.34502840170047, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 118.887657
 - $\hat{q}^\beta$: 118.887657
 - q^β : 118.887657
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 131.69760457629607, 147.48374761404892, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 2.545035
 - $\hat{q}^\beta$: 2.545035
 - q^β : 2.545035
 - Breakpoints: [0, 2.9819181007907165, 3.1809419553840232, 4.432280503220532, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 38.216522
 - $\hat{q}^\beta$: 38.216522
 - q^β : 38.216522

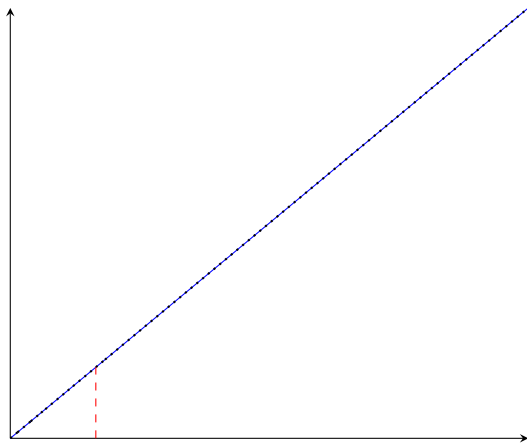
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, **37.47875108132638**, **38.88483572219945**, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.468286
 - $\hat{q}^\beta$: 13.468286
 - q^β : 13.468286
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 13.192252946941991, 13.22717920580809, 13.244985069124171, **13.30871858348944**, **14.09178614751222**, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



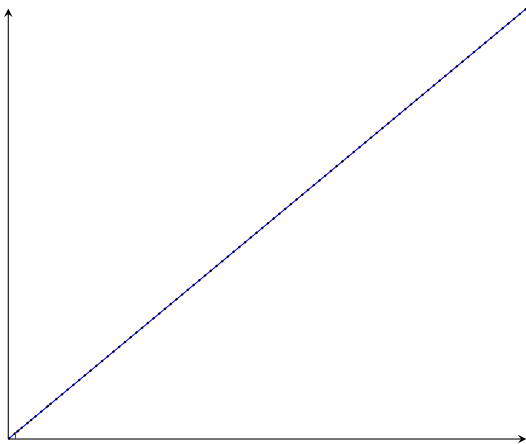
(a) $q^\beta - (1, 1, 1)$



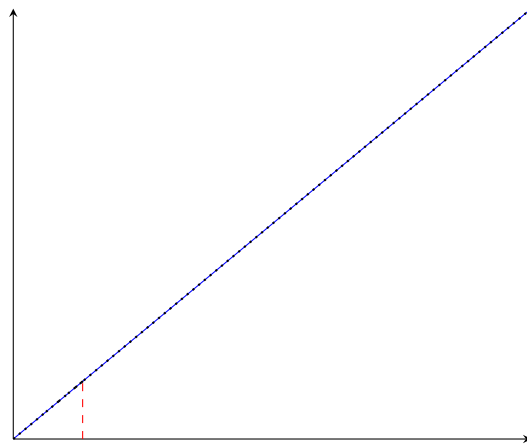
(b) $q^\beta - (1, 2, 2)$



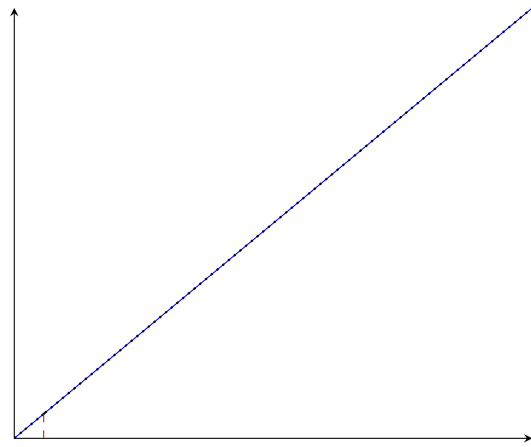
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	5456.856442255253
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1976.248991364448
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.008164772846778379	$b_{2,1,1}^{C,in}$	260.29705449980867
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	1976.2489913644486
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	260.29705449980867
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	418.8196134269507		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	240.0	$t_{2,2,2}^H$	523.7108896292779		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	11.849331378965468	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	421.74597685443155		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4821.204716535508		
$f_{1,2,1}^H$	7.417238025439243	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5456.856442255253		
$f_{2,2,2}^H$	9.248909224161444	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.8201194911328	$b_{1,2,1}^{H,out}$	4821.204716535509		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.4607143943927685	$b_{2,2,1}^{H,out}$	0.0		